ANGAD SINGH BHATTI

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EDUCATION

Fordham University, Gabelli School of Business — New York, NY
MS, Quantitative Finance — May 2024 – May 2026
GPA: 3.75/4.00 (Dean's Scholarship)

- Stochastic Calc, Comp Finance, Adv Credit Risk Modeling, Crypto Algotrading, Market Microstructure, Cloud Computing

Pune Institute of Computer Technology (SPPU) — Pune, India
Bachelor of Engineering, Computer Science — August 2018 – August 2022
GPA: 3.53/4.00

- Graduated with honors in Data Science with distinction (4.00/4.00 in Data Science Honors)

RESEARCH EXPERIENCE

Research on Fractional Differencing with Prof. Dmitry Udler — September 2025 – Present

- Developed estimator of long memory addressing failure of standard ARFIMA methods (EW collapses on market data)
- Outperformed fixed $d = 0.5$ in **80%** of assets on residual whitening, and dominated it on ML forecasts in low-memory regimes, in a walk-forward test across S&P 500 constituents (2010–2025)
- Documented return skewness deepening from **-0.14** to **-0.46** across memory regimes
- 2 SSRN working papers drafted (Market Memory, Persistence Estimation); Accepted in peer reviewed conf: ITISE 2026
- Showed memory regimes predict forward 21-day realized skew beyond model-free implied skew across 4 equities (Research Note)

Research and Teaching Assistant with Prof. Ren Raw Chen — January 2025 – May 2026

- Modeled 13F data (4,634 funds, 4,824 assets, 30 years) as a bipartite network and constructed graph embeddings using node2vec
- Measured latent fund similarity to study liquidity clustering and potential fire-sale contagion channels among funds
- Teaching assistant for 2 graduate level classes: Financial Theory and AI in Asset Management

Applied Econometrics Project, sponsored by AutoCase Economic Advisory — September 2025 – December 2025

- Computed WTP from survey data, correcting for sampling bias and modeling trust in government as a latent variable via an Ordered Probit Selection model with DAG-derived causal features, improving model R^2 from 0.08 to 0.23 (Canadian estuary)

EXPERIENCE

Rebellion Research — Remote, NY
Quant Research Intern (Python) — June 2025 – September 2025

- Studied 13F data to analyze common holdings of portfolio managers to build internal report and trading strategy
- Authored and published an interview with Deutsche Bank's Global Quant Head on the evolution of the QIS space

Pre-Grad Study — May 2023 – May 2024

- Self-study in Real Analysis, Probability Theory, Linear Algebra, Stochastic Processes, Asymptotic Theory

FlexTrade — Pune, India
Associate Software Developer (C++) — February 2023 – May 2023

- Merged market-making (scheduling) and FVPEG (pricing) into a unified pipeline (8000 LOC)
- Managed order states and optimized the merged code base for latency

Interactive Brokers — Mumbai, India
Junior Software Engineer (Java) — August 2022 – December 2022

- Maintained and extended trade-matching engine functionality in Java through bug fixes, edge-case handling
- Contributed to a multithreaded platform designed to support heterogeneous report formats through a scalable architecture

PROJECTS

Signal Incrementality Analysis for Revenue Forecasting — June 2026 – June 2026

- Evaluated retail-sales data as a leading indicator of Walmart revenue using a walk-forward forecasting framework; achieved a 14% reduction in out-of-sample RMSE versus a naive seasonal baseline (writeup)

Carver Trend-Following Implementation — May 2025 – October 2025

- Implemented risk targeting strategies from Carver's textbook; Strategy 5 attained Sharpe ratio of **1.2** after 10 bps tc over 2005–2025 (walk-forward, 35 diversified futures; consistent on crypto portfolio) (writeup)

SKILLS

Programming: Python, C++, Java; NumPy, Pandas, SciPy, statsmodels, scikit-learn, TensorFlow

Methods: Time-Series Econ., Stochastic Calc., Statistical Arb., Causal Inference, Monte Carlo Simulation, Computer Vision

Tools: Git, Linux/Bash, SQL, AWS, Multithreading, OOP, Data Structures and Algorithms

Other Interests: Chess (FIDE 1500); Gym; Reading; mental maths(UCMAS certified); MUNs